

1.

A. The following is a dump of a UDP header in hexadecimal format.

0045DF000058FE20

a) **What is the source port number?**

The source port number is the first 2 bytes (4 hexadecimal digits). In this case, it is 0045, which is 69 in decimal.

b) **What is the destination port number?**

The destination port number is the next 2 bytes (4 hexadecimal digits). In this case, it is DF00, which is 57088 in decimal.

c) **What is the total length of the user datagram?**

The total length of the user datagram is the next 2 bytes (4 hexadecimal digits). In this case, it is 0058, which is 88 in decimal.

d) **What is the length of the data?**

The length of the data can be calculated by subtracting the UDP header length (8 bytes) from the total length. So, 88 (total length) - 8 (UDP header length) = 80 bytes.

e) **Is the packet directed from a client to a server or vice versa?**

The message is from a server with small (well-known) port number to a client with a large (ephemeral) port number.

f) **What is the client process?**

The well-known port number 69 belongs to TFTP.

B. **How do opening, closing and shrinking of window occur in TCP? Give examples for both send receive window.**

Opening of Windows:

- **Sender Window (Send Window):** Starts small and increases as acknowledgments are received.
 - ✓ **Example:** Initially set to 10 segments, may increase to 15 as acknowledgments arrive.
- **Receiver Window (Receive Window):** Represents buffer space; increases as data is processed.
 - ✓ **Example:** Initially set to 20 segments, may increase to 25 as data is acknowledged.

Closing of Windows:

- **Sender Window (Send Window):** May close or shrink due to congestion or network issues.
 - ✓ **Example:** Reduces from 20 to 15 during congestion.
- **Receiver Window (Receive Window):** Closes when the receiver's buffer is full.
 - ✓ **Example:** Closes when it has acknowledged 30 packets and needs to process them.

C. what are the services that a transport-layer protocol can offer to applications invoking it? Describe each with example.

- **End-to-End Delivery:** Ensuring data is reliably delivered from the source to the destination.
 - ✓ **Example:** TCP guarantees end-to-end delivery of web pages, emails, and files by retransmitting lost packets.
- **Addressing:** Identifying the source and destination of data using addresses or ports.
 - ✓ **Example:** In TCP/IP, source and destination IP addresses and port numbers are used to address data packets.
- **Reliable Delivery:** Ensuring data arrives without errors and in the correct order.
 - ✓ **Example:** TCP provides reliable delivery by using acknowledgments and retransmissions for error-free file transfers.
- **Flow Control:** Managing the rate of data transmission to prevent congestion and ensure efficient delivery.
 - ✓ **Example:** TCP employs flow control mechanisms like windowing to regulate data flow between sender and receiver.
- **Multiplexing:** Supporting multiple applications on the same network connection.
 - ✓ **Example:** TCP and UDP use port numbers to multiplex data, allowing multiple applications like web browsers and email clients to use a single network link simultaneously.

D. A sender sends a JPEG message. Show the MIME header.

The MIME (Multipurpose Internet Mail Extensions) header for a JPEG message typically looks like this:

Content-Type: image/jpeg

This header indicates that the content of the message is in JPEG format.

- A. Suppose, An FTP server has received a packet from an FTP client with IP address 153.2.7.9. The FTP server wants to verify that the FTP client is an authorized client. The FTP server can consult a file containing the list of authorized clients. However, the file consists only of domain names. The FTP server has only the IP address of the requesting client, which was the source IP address in the received IP datagram. The FTP server asks the resolver (DNS client) to send an inverse query to a DNS server to ask for the name of the FTP client. Show the query and response messages with values for each field. Discuss separately.

0x1200				0x0900			
1				0			
0				0			
1	'9'	1	'7'	1	'7'	1	'7'
1	'2'	3	'1'	3	'1'	3	'1'
'5'	'3'	7	'i'	7	'i'	7	'i'
'n'	'-'	'a'	'd'	'a'	'd'	'a'	'd'
'd'	'r'	4	'a'	4	'a'	4	'a'
'r'	'p'	'a'	0	'a'	0	'a'	0
12				1			

The first 2 bytes show the identifier (0x1200). The flags value is 0x0900 in hexadecimal. In binary it is 0000100100000000, and we divide it into fields as shown below:

QR	OpCode	AA	TC	RD	RA	Reserved	rCode
0	0001	0	0	1	0	000	0000

The OpCode is 0001, which defines an inverse query. The message contains only one question record. The domain name is 19171231537in-addr4arpa. The next 2 bytes define the query type as PTR, and the last 2 bytes define the class as the Internet. Figure 19.24 shows the response. The flags value is 0x8D80 in hexadecimal. In binary it is 1000110110000000, and we divide it into fields as shown below:

QR	OpCode	AA	TC	RD	RA	Reserved	rCode
1	0001	1	0	1	0	000	0000

0x1200		0x8D80	
1		1	
0		0	
1	'9'	1	'7'
1	'2'	3	'1'
'5'	'3'	7	'i'
'n'	'-'	'a'	'd'
'd'	'r'	4	'a'
'r'	'p'	'a'	0
12		1	
0xC00C		12	
1		Continued on next line	
24000		10	
4	'm'	'h'	'h'
'e'	3	'c'	'o'
'm'	0		

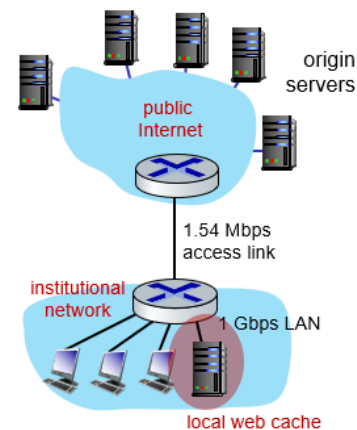
B. Describe how web caching can reduce the delay in receiving a requested object. Will web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

Calculating access link utilization, end-end delay with cache:

suppose cache hit rate is 0.4:

- 40% requests served by cache, with low (msec) delay
- 60% requests satisfied at origin
 - rate to browsers over access link
 $= 0.6 * 1.50 \text{ Mbps} = .9 \text{ Mbps}$
 - access link utilization $= 0.9 / 1.54 = .58$ means low (msec) queueing delay at access link
- average end-end delay:
 - $= 0.6 * (\text{delay from origin servers})$
 - $+ 0.4 * (\text{delay when satisfied at cache})$
 - $= 0.6 (2.01) + 0.4 (\sim \text{msecs}) = \sim 1.2 \text{ secs}$

lower average end-end delay than with 154 Mbps link (and cheaper too!)



Web caching can reduce the delay for all objects, even objects that are not cached, since caching reduces the traffic on links.

C. What are the services provided by the SSH protocol through port forwarding? Explain with example.

- **Local Port Forwarding:** Redirects traffic from a local port on the client to a remote server.
 - ✓ **Example:** Forwarding local port 8080 to a remote web server's port 80. This allows accessing the remote web server through **localhost:8080** on the client.
- **Remote Port Forwarding:** Redirects traffic from a remote port on the server to a local client.
 - ✓ **Example:** Forwarding remote port 3306 on the server to the client's port 3306. This allows the server to connect to a database running on the client.
- **Dynamic Port Forwarding (SOCKS Proxy):** Creates a dynamic SOCKS proxy on the client, allowing multiple connections to be forwarded through a single SSH tunnel.
 - ✓ **Example:** Establishing a dynamic port forwarding tunnel and configuring a web browser to use it as a SOCKS proxy. This enables secure and anonymous web browsing through the SSH tunnel.

D. Show the encoding for the INTEGER 1456 in case of SNMP. (Chapter-28)

3.

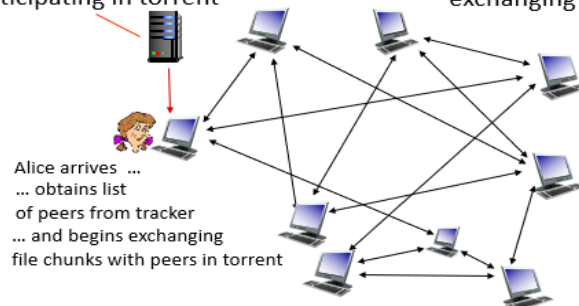
A. Suppose a content provider, NetCinema, employs the third-party CDN company, KingCDN, to distribute its videos to its customers. On the NetCinema web pages, each of its videos is assigned a URL that includes the string "video" and a unique identifier for the video itself; for example, Transformer 7 might be assigned <http://video.netcinema.com/6Y7B23V> . If DASH is used , what will be the steps before the client can dynamically select chunks from the different versions of the video?(Chapter-25).

B. Consider a new peer Alice that joins BitTorrent without possessing any chunks without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk?

- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



Recall that in BitTorrent, a peer picks a random peer and 'optimistically unchokes' the peer for a short period of time. Therefore, Alice will eventually be optimistically unchoked by one of her neighbors, during which time she will receive chunks from that neighbor.

C. Define Jitter. Explain the application to TFTP in conjunction with DHCP.

Jitter is the variation in the delay of received packets in a network, particularly in packet-switched networks like the Internet.

In conjunction with DHCP (Dynamic Host Configuration Protocol), TFTP (Trivial File Transfer Protocol) is often used during the boot process of networked devices, such as computers and network appliances, to retrieve configuration or boot files. Here's a brief explanation of their application together:

- **DHCP Configuration:** DHCP dynamically assigns IP addresses and network configuration settings to devices when they connect to a network. This includes information like the device's IP address, subnet mask, default gateway, and DNS server addresses.
- **TFTP Boot:** In some cases, DHCP can also provide information about a TFTP server and a specific file to retrieve. This is particularly useful in scenarios where devices need to boot from a network, such as thin clients, network-attached storage (NAS) devices, or diskless workstations.
- **TFTP File Retrieval:** The device, upon receiving the DHCP-assigned information, knows that it should retrieve a specific file from the TFTP server mentioned in the DHCP response. It uses TFTP to request and download this file from the server.
- **File Examples:** The file retrieved via TFTP could be various things, such as a boot loader, an operating system kernel, or configuration files needed for the device to function properly within the network.

D. Do you think H.323 is actually the same as SIP? Make a comparison between the two.

No, H.323 and SIP (Session Initiation Protocol) are not the same; they are two different protocols used for real-time communication over the Internet and IP-based networks.

Here's a comparison between the two:

1. H.323 protocol is based on telephony while SIP is based on internet.
2. The H.323 standard body is ITU. In contrast, SIP is designed by IETF.
3. H.323 follows binary message format. Conversely, SIP uses ASCII message format.
4. H.323 is not compatible with internet while SIP has the compatibility with the internet.
5. The architecture of H.323 is monolithic. Inversely, SIP is built on the modular architecture.
6. SIP offers the instant messaging facility. On the contrary, there is no such facility in H.323.